

3D Kinetic Energy Harvesting Shoe Sole

2025 FUSD and MSJ Rotary 3D Printing Maker Faire

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Harvesting Shoe Sole

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Instructables.com

<https://www.instructables.com/Piezoelectric-Shoe-Soles/>

Link:

QR Code:



Problem Statement

- Our planet faces an **energy crisis** due to climate change and pollution.
- **Traditional energy** sources contribute to greenhouse gases and are unsustainable.
- **Kinetic energy** from human movement is wasted every day.
- **Sedentary lifestyles** contribute to health problems like obesity and heart disease.

Mission

- **Integrates** piezoelectric sensors into 3D-printed shoe soles to harvest energy.
- **Converts** mechanical energy from walking into electrical energy.
- Energy can be **stored** in batteries or capacitors to power devices like fitness trackers and smartphones.
- **Encourages** sustainable energy generation and **reduces** reliance on fossil fuels.

Vision

- Create a **sustainable and healthy future** through kinetic energy harvesting.
- Develop energy-harvesting shoes as **commercially viable products**.
- Imagine **cities powered by kinetic energy** from walking citizens.
- Integrate a **smart tracking system** for real-time energy monitoring and motivation.
- Combine energy sustainability with **personal fitness** for global impact.

Rotary Values

Protecting Environment



- Reduces reliance on fossil fuels
- Converts wasted kinetic energy into usable electricity

Promoting Lifestyles



- Promotes healthier lifestyles by motivating people to walk and exercise more and generate energy.

Supporting Education

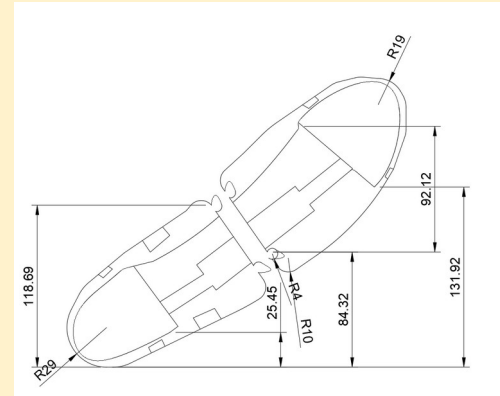
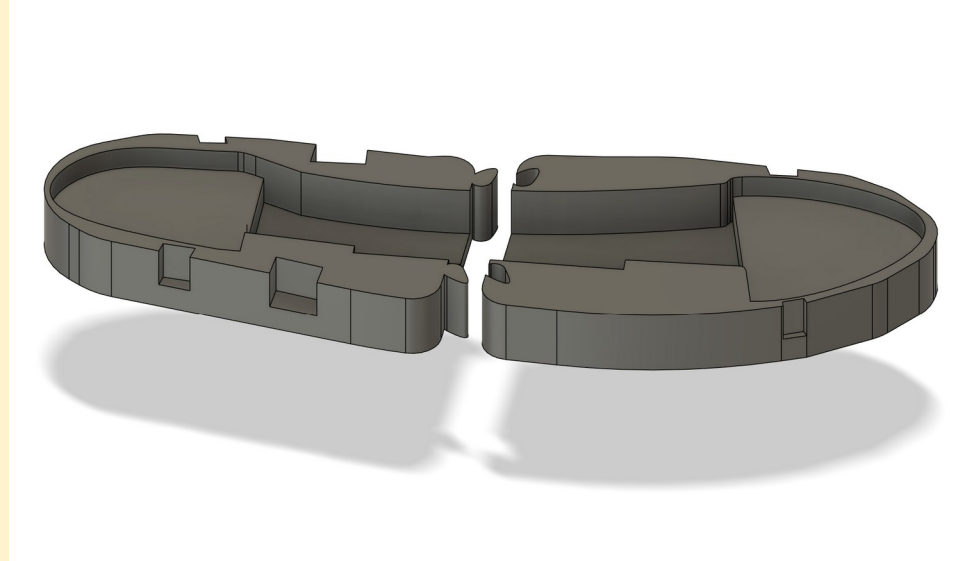


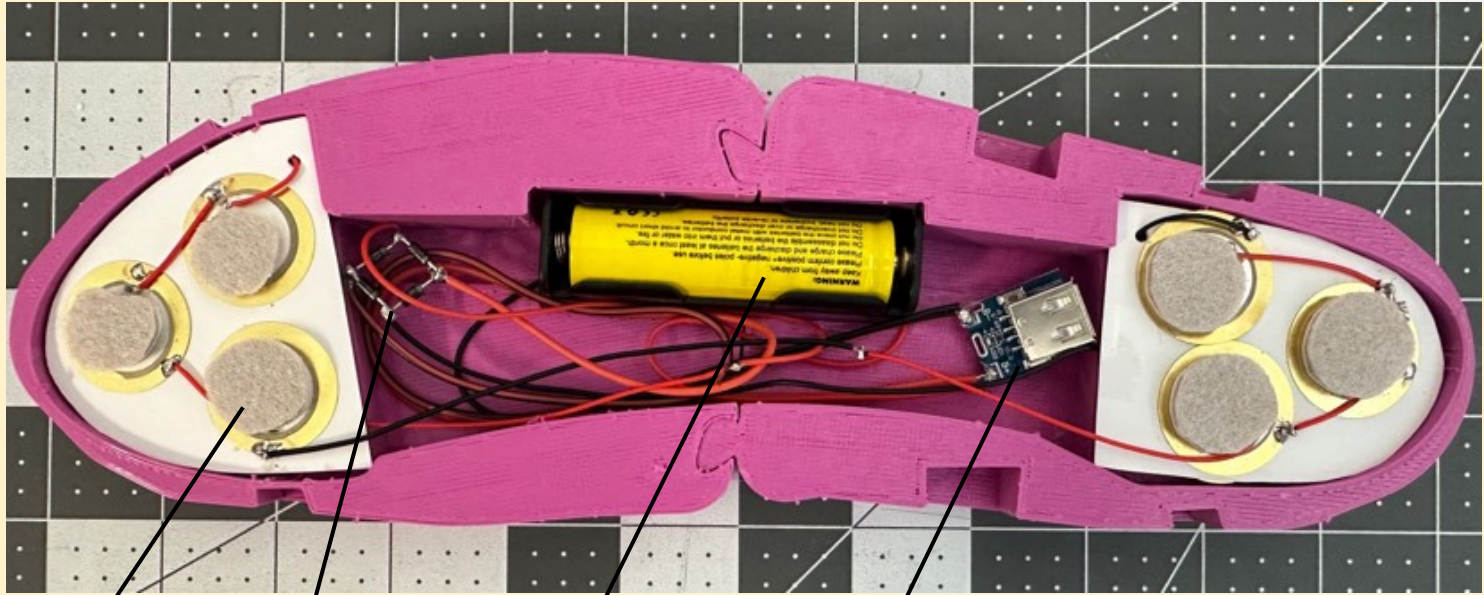
- Provides an excellent tool for teaching renewable energy and sustainability.
- Encourages greener living and raises awareness of personal energy consumption.

Project Summary

- Our project focuses on developing sustainable kinetic energy harvesting shoes by integrating piezoelectric sensors with 3D technology.
- Our project utilizes a 3D-printed shoe sole to house the electrical components.
- The piezoelectric sensors generate power when people walk or run.
- The generated power can be stored in an embedded battery to charge wearable devices and cell phones.
- NinjaFlex 3D Printer Filament (85A) housing ensures comfort and durability.

Shoe Sole
3D Design:



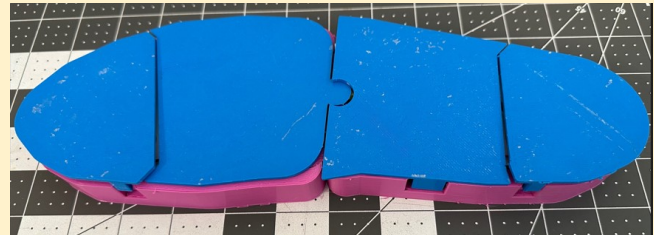


**Power from
Piezoelectric Discs**

**Energy Storage
(Battery)**

**Charging Module
(USB Output)**

Diode Rectifier



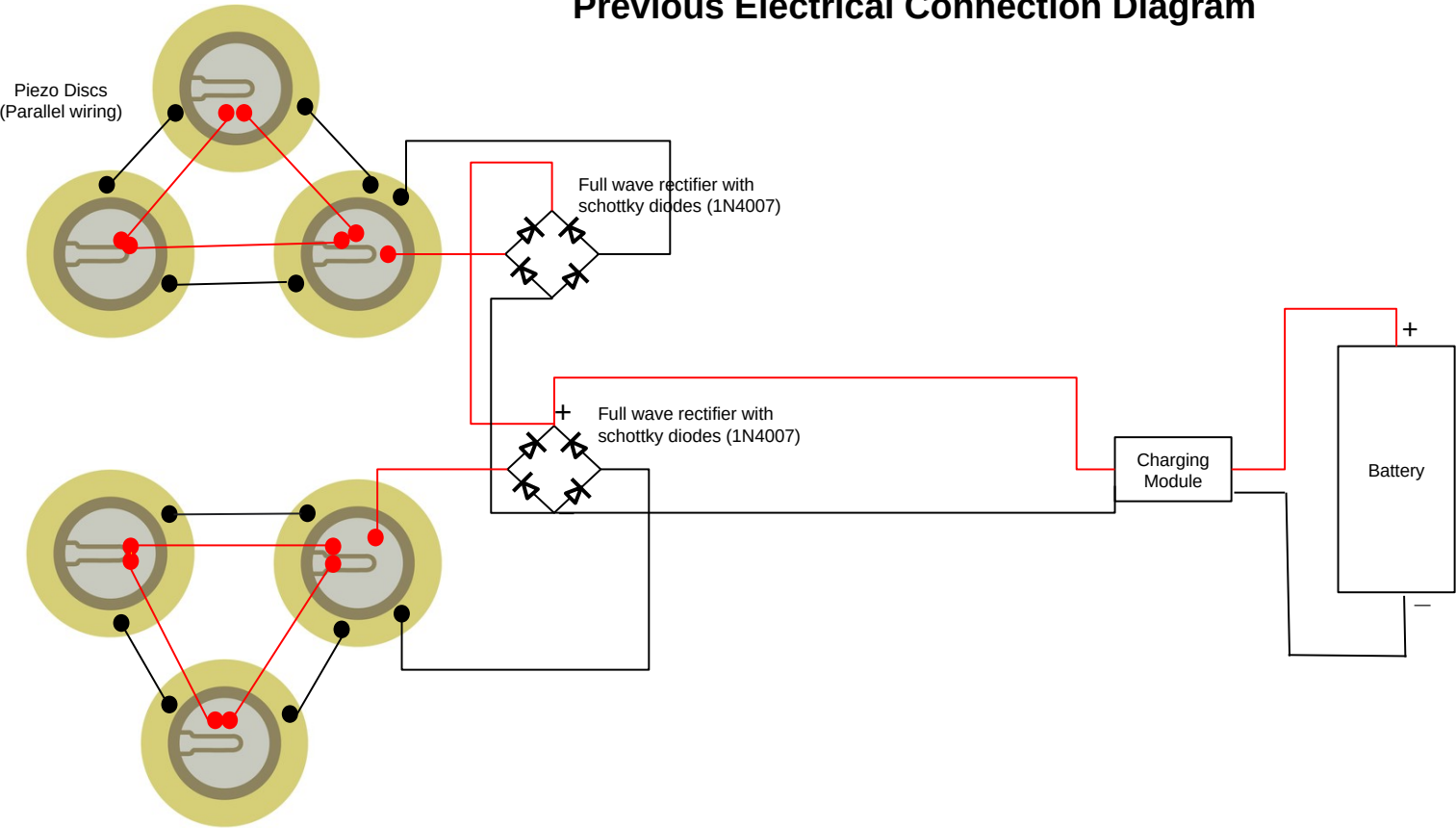
Electrical Components

- 12 x 27mm Piezoelectric Discs
- 4 x Schottky Diodes (1N5819) for a Bridge Rectifier
- 5V Boost Converter (Power Supply Module Lithium Battery Li-ion Charging Protection Board 134N3P)
- 18650 Battery Storage Case/Charging Board
- 18650 Rechargeable Battery

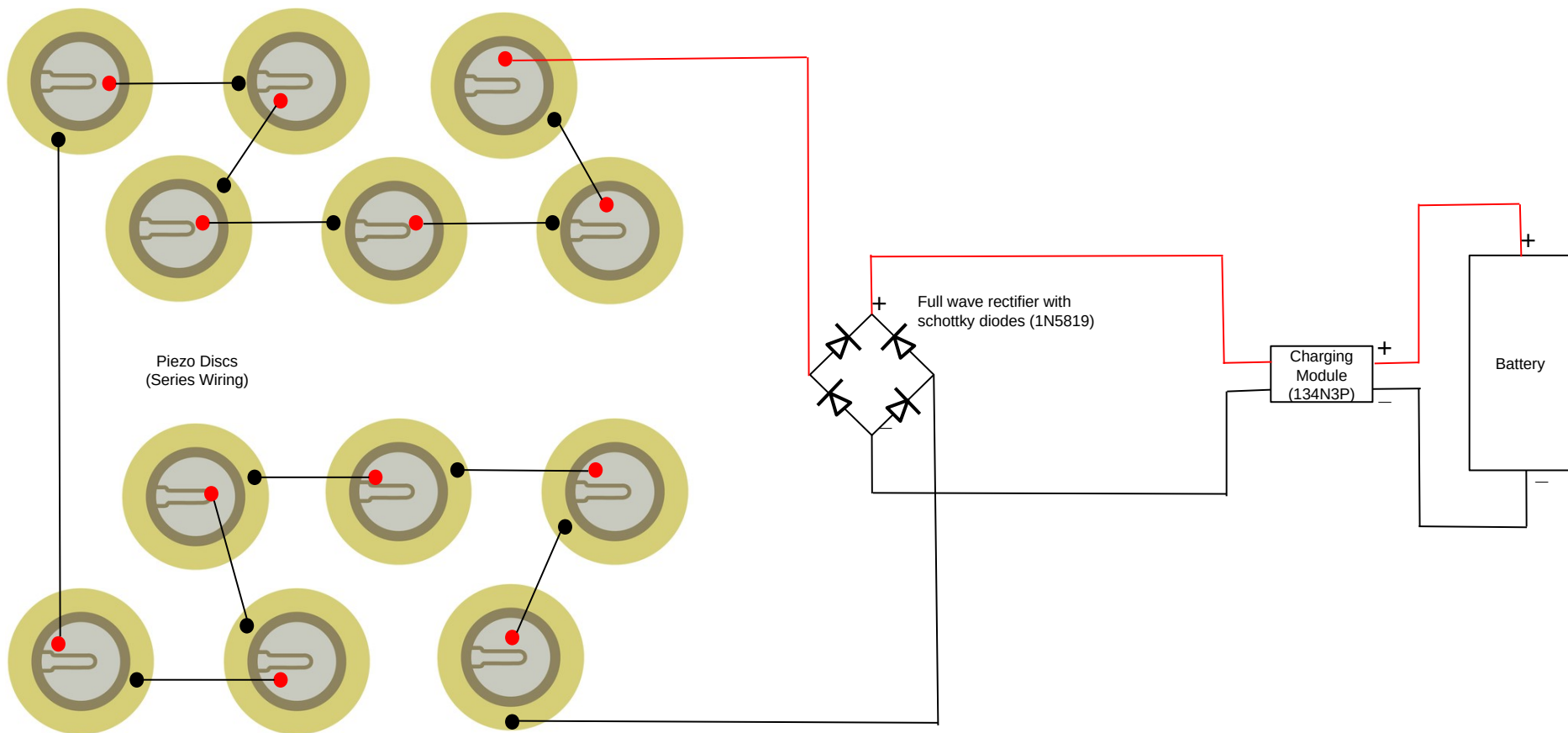
Electrical Component Connections

1. **Connect Piezo Discs in Series:**
 - Take **Piezo Disc #1** and connect its **negative (-) lead** to the **positive (+) lead of Piezo Disc #2**.
 - Repeat the wiring pattern on the rest **Piezo Discs**.
 - The remaining **positive lead** of Piezo Disc #1 and the **negative lead** of Piezo Disc #12 will be the final output.
1. **Connect to a Bridge Rectifier:**
 - **Final Positive Lead (Piezo #1)** → Connect to **AC1** input of the bridge rectifier.
 - **Final Negative Lead (Piezo #12)** → Connect to **AC2** input of the bridge rectifier.
1. **Rectify AC to DC Using Schottky Diode Bridge:**
 - **Bridge Rectifier DC Output (+)** → Connect to **Boost Converter IN+**.
 - **Bridge Rectifier DC Output (-)** → Connect to **Boost Converter IN-**.
1. **Charge the Battery:**
 - **Boost Converter 5V OUT+** → Connect to **Battery Charging Board IN+**.
 - **Boost Converter Ground** → Connect to **Battery Charging Board IN-**.
1. **Connect to 18650 Battery:**
 - **Battery Charging Board Output** → Connect to **Battery Holder + and -**.

Previous Electrical Connection Diagram



Improved (Current) Electrical Connection Diagram



Improvements of the Electrical Circuit

1. Increased Piezoelectric Discs from 6 to 12:

- **Higher Energy Output:** Increased the total harvested kinetic energy, leading to improved efficiency.

2. Changed Wiring from Parallel to Series:

- **Voltage Boost:** Series configuration noticeably increases voltage output
- **Efficiency Improvement:** Reduces current losses, enhancing overall efficiency of energy conversion.

3. Replaced 1N4007 Diodes with 1N5819 Schottky Diodes:

- **Reduced Voltage Drop:** Lower forward voltage drop (0.3-0.4V vs. 0.7-1.0V) increases efficiency.
- **Faster Switching Speeds:** Allows more efficient rectification, improving energy harvesting efficiency.

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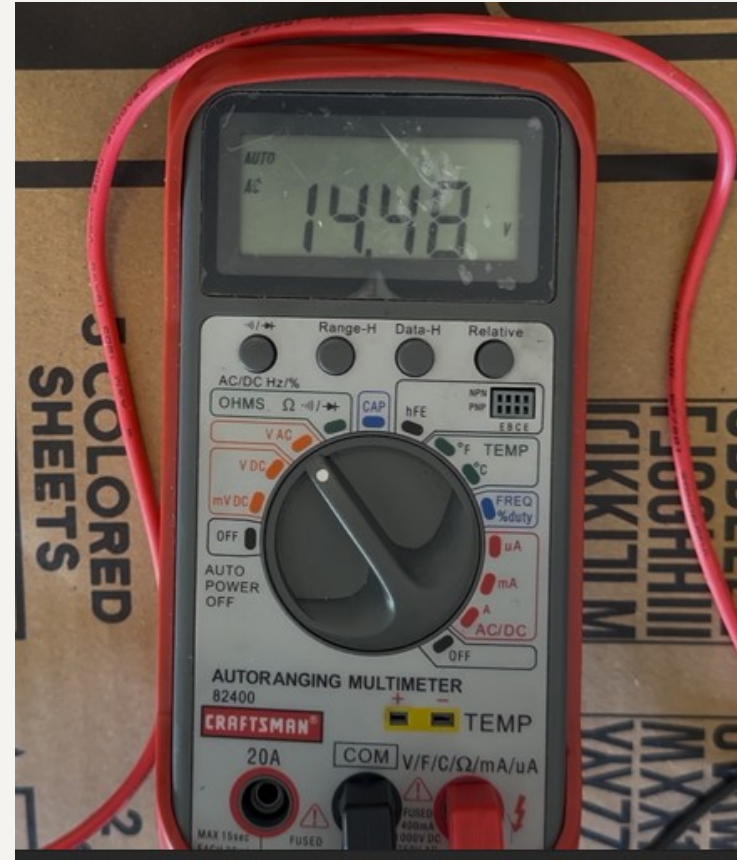
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Experiment #1: Open-Circuit Voltage Measurement

- **Objective:** Measure peak voltage generated by the piezoelectric discs.
- **Procedure:**
 - Connect the piezoelectric discs (series configuration) to the multimeter set to AC voltage mode.
 - Set multimeter to AC Voltage mode.
 - Apply repeated pressure or simulate footsteps at our assumed frequency (~1.5 steps/sec).
 - The maximum voltage measured:
14.48V



Experiment #2: Rectified DC Voltage Measurement

- **Objective:** Measure voltage after the diode bridge rectifier.
- **Procedure:**
 - Connect the multimeter to the output of the diode bridge.
 - Repeat simulated footsteps; measure and record the stabilized DC voltage.
 - The maximum voltage measured:
9.65V



Experiment #3:

Energy Accumulation and Battery Charging Test

- **Objective:** Determine the energy the battery stored during the 30-minute walk by taking the 3D shoe sole.
- **Procedure:**
 - Set the multimeter to DC Voltage mode.
 - Measure and record battery voltage before the experiment (3.81V).
 - Simulate walking for a fixed duration (e.g., 30 mins) to generate energy.
 - Immediately after a 30-minute walk, take the battery out of the battery holder, measure and record battery voltage after the walk (3.823V).
 - Connect a 220 Ω resistor across battery terminals to start a discharge test.
 - Measure and record battery voltage every 10 minutes during discharge until reaching our original voltage (3.81V).
 - Calculate the energy stored by integrating power ($P = V^2/R$) over time ($E=P \times t$).



Calculations in Experiment #3

❖ Joule's Law (Power Equation):

$$P = V^2 / R$$

- **P:** Power (Watts)
- **V:** Voltage (Volts)
- **R:** Resistance (Ohms)

E.g., $P = (3.817V)^2 / 220\Omega = 0.0662W$

❖ Energy: $E = P \times t = (V^2 / R) \times t$

- **E:** Energy in Joules
- **t:** Time in seconds per interval (600 sec = 10 min)

E.g., $E = 0.0662W \times 600s = 39.72Joules$

❖ Energy Over Time:

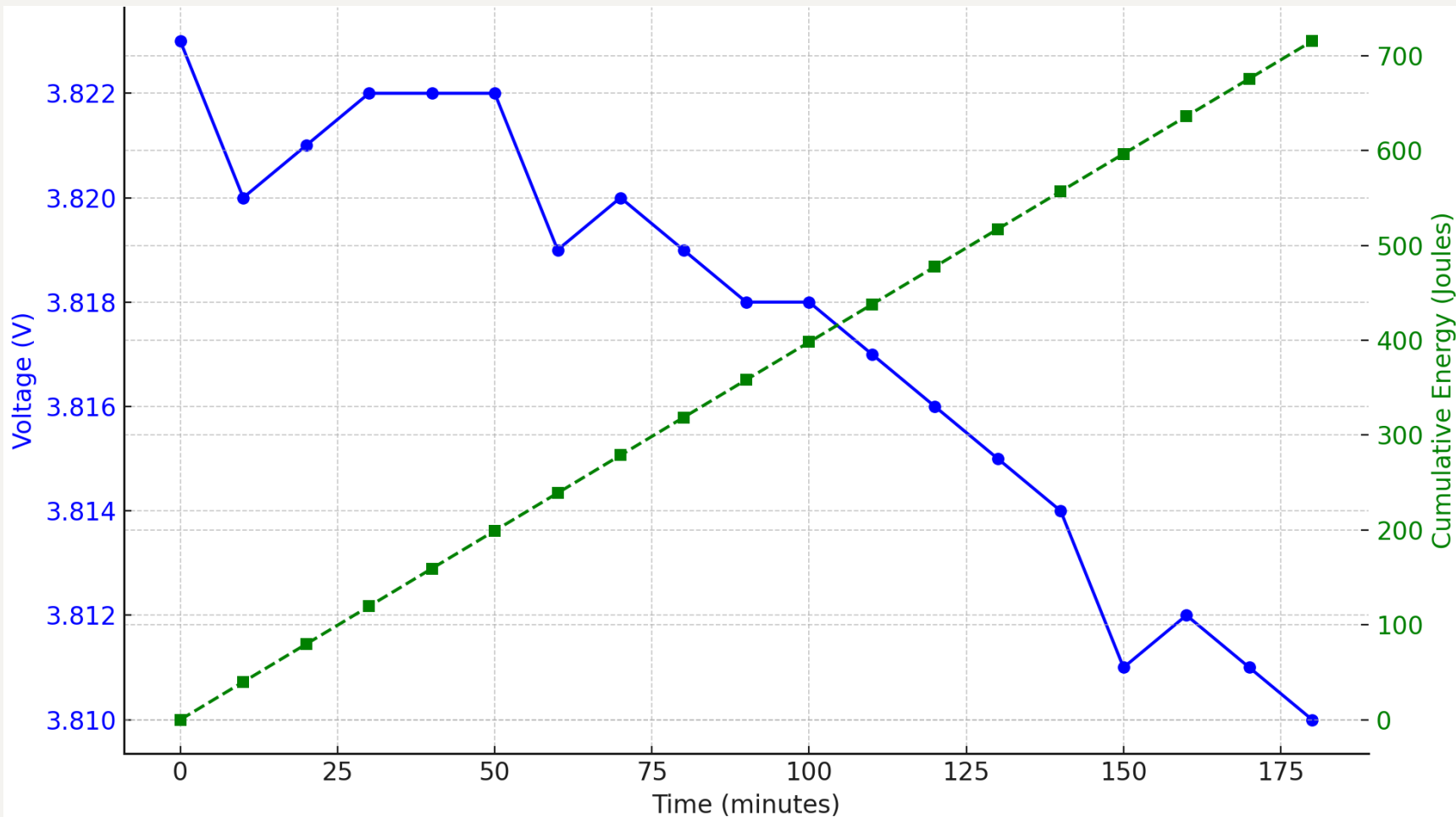
Integration of power over time gives the total energy (in joules).

- ❖ **Note:** In our 3D piezoelectric shoe sole experiment, the current output is extremely low. These tiny currents are too small to be reliably measured with standard multimeters during discharge. Therefore, we can't directly use **$I = V/R$ (Ohm's Law)** because **I** (current) is not directly measurable (**I**: Current (Amps), **V**: Voltage (Volts), **R**: Resistance (Ohms)).

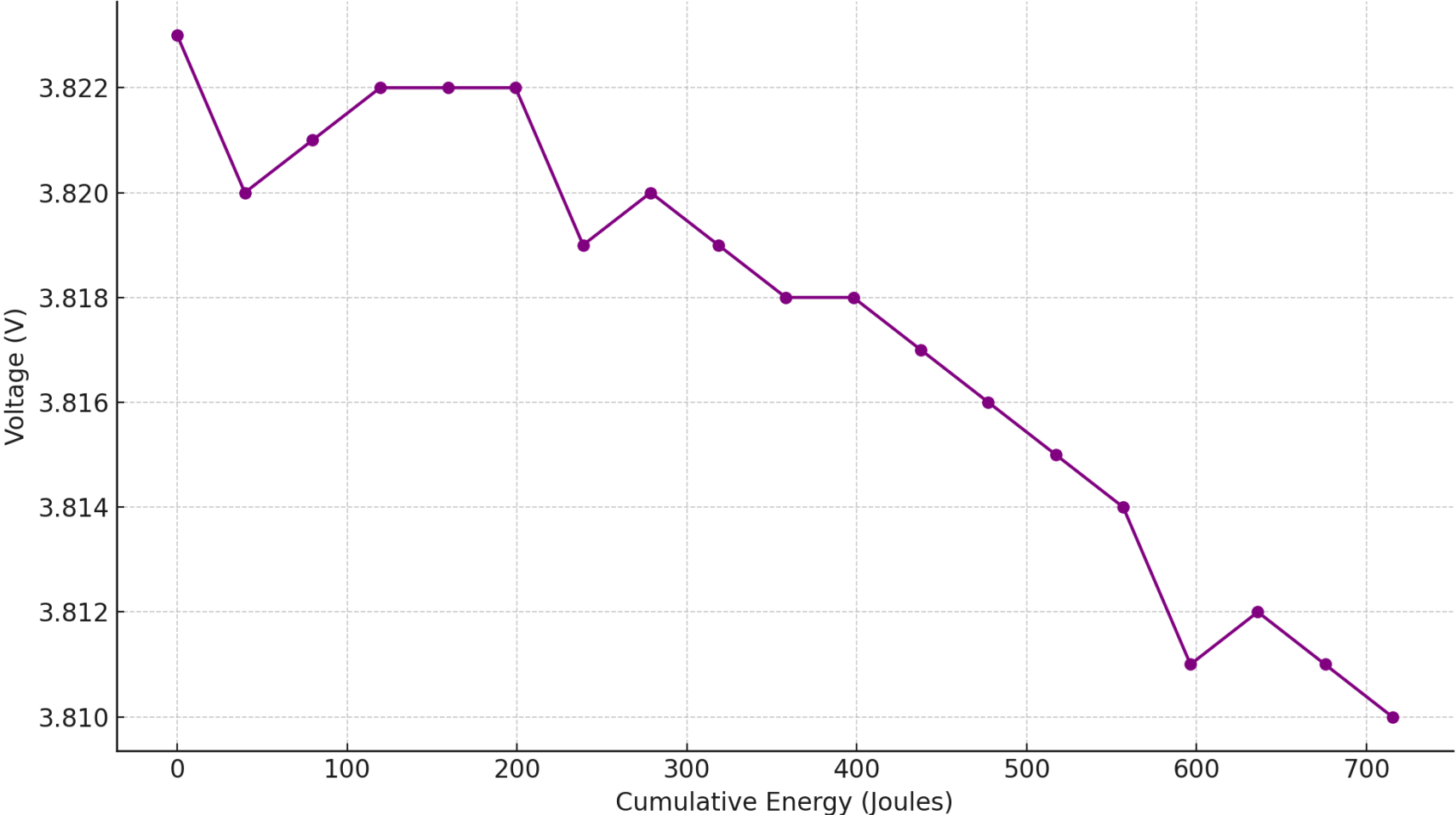
Battery Discharge Experiment 1

	Time (min)	Voltage (V)	Voltage ² (V ²)	Power (W)	Energy per Interval (J)	Cumulative Energy (J)
Post-Walk Voltage	0	3.823	14.615329	0.06643331364	0	0
	10	3.82	14.5924	0.06632909091	39.79745	39.79745
	20	3.821	14.600041	0.06636382273	39.81829	79.61574
	30	3.822	14.607684	0.06639856364	39.83914	119.45488
	40	3.822	14.607684	0.06639856364	39.83914	159.29402
	50	3.822	14.607684	0.06639856364	39.83914	199.13316
	60	3.819	14.584761	0.06629436818	39.77662	238.90978
	70	3.82	14.5924	0.06632909091	39.79745	278.70723
	80	3.819	14.584761	0.06629436818	39.77662	318.48385
	90	3.818	14.577124	0.06625965455	39.75579	358.23964
	100	3.818	14.577124	0.06625965455	39.75579	397.99543
	110	3.817	14.569489	0.06622495	39.73497	437.7304
	120	3.816	14.561856	0.06619025455	39.71415	477.44455
	130	3.815	14.554225	0.06615556818	39.69334	517.13789
	140	3.814	14.546596	0.06612089091	39.67253	556.81042

Battery Discharge Test 1: Voltage & Cumulative Energy vs. Time



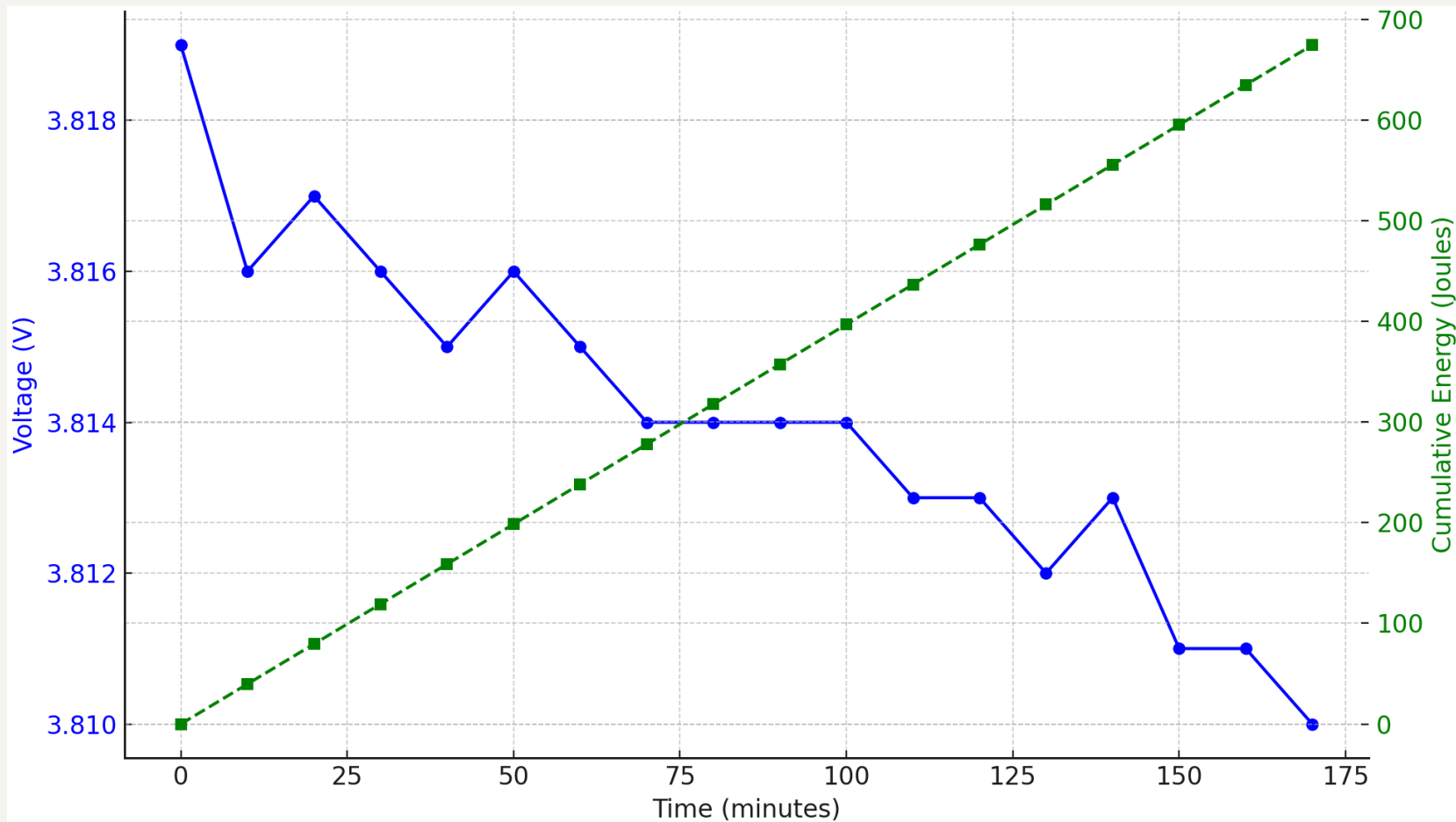
Battery Discharge Test 1: Voltage vs. Cumulative Energy



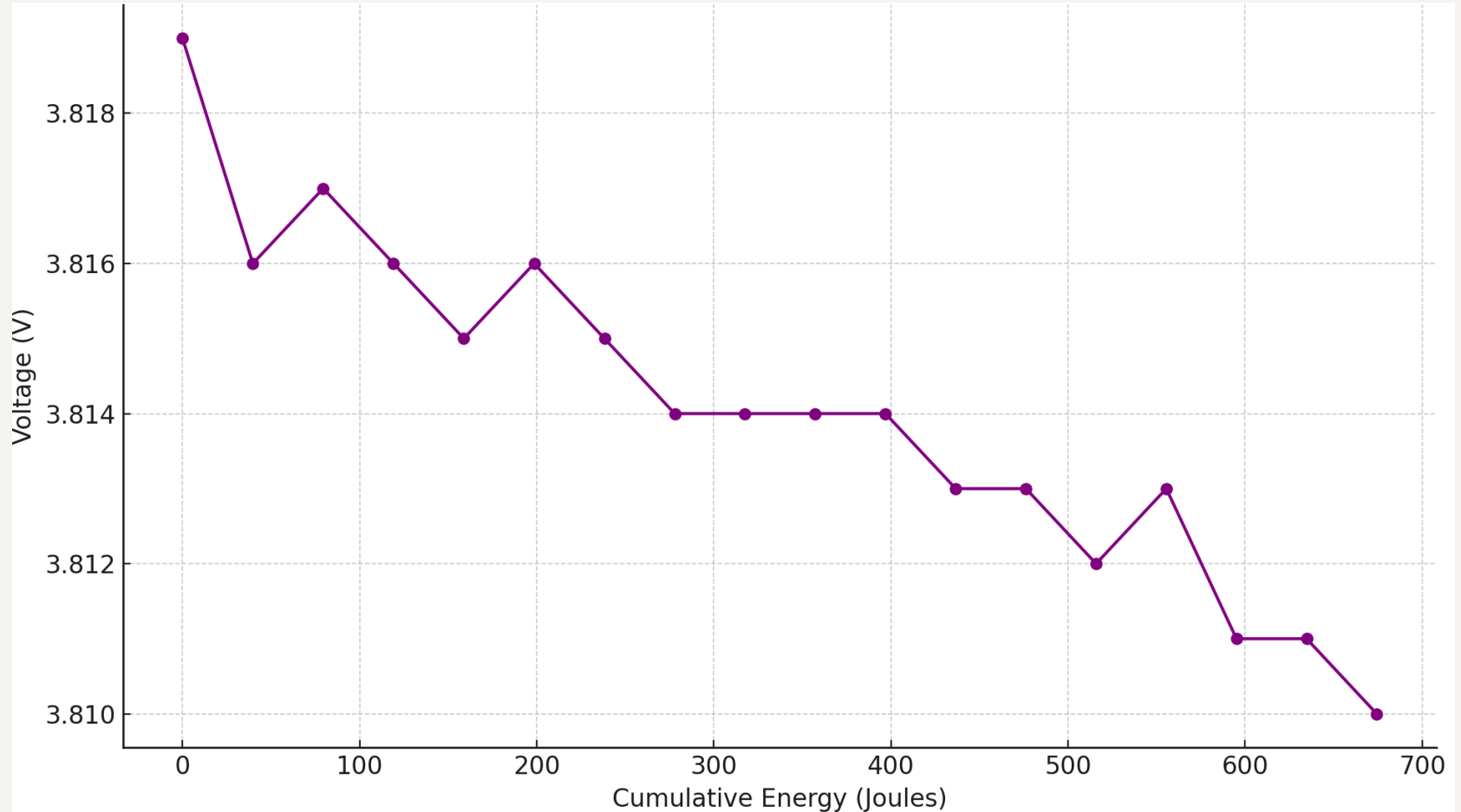
Battery Discharge Experiment 2

	Time (min)	Voltage (V)	Voltage ² (V ²)	Power (W)	Energy per Interval (J)	Cumulative Energy (J)
Post-Walk Voltage	0	3.819	14.584761	0.06629436818	0	0
	10	3.816	14.561856	0.06619025455	39.71415	39.71415
	20	3.817	14.569489	0.06622495	39.73497	79.44912
	30	3.816	14.561856	0.06619025455	39.71415	119.16327
	40	3.815	14.554225	0.06615556818	39.69334	158.85661
	50	3.816	14.561856	0.06619025455	39.71415	198.57076
	60	3.815	14.554225	0.06615556818	39.69334	238.2641
	70	3.814	14.546596	0.06612089091	39.67253	277.93663
	80	3.814	14.546596	0.06612089091	39.67253	317.60916
	90	3.814	14.546596	0.06612089091	39.67253	357.28169
	100	3.814	14.546596	0.06612089091	39.67253	396.95422
	110	3.813	14.538969	0.06608622273	39.65173	436.60595
	120	3.813	14.538969	0.06608622273	39.65173	476.25768
	130	3.812	14.531344	0.06605156364	39.63094	515.88862

Battery Discharge Test 2: Voltage & Cumulative Energy vs. Time



Battery Discharge Test 2: Voltage vs. Cumulative Energy



Experimental Conclusion

Experiment #1: Open-Circuit Voltage Measurement

- Max Voltage Output (AC, no load): 14.48 V
- Confirms that the piezoelectric discs can generate high instantaneous voltage under footstep simulation

Experiment #2: Rectified DC Voltage Measurement

- Max Voltage After Bridge Rectifier: 9.65 V
- Voltage drop observed due to diode losses (Schottky diode bridge)
- Confirms AC-to-DC conversion works, with expected diode voltage drop (~4.8V total)

Experiment #3: Battery Charging & Energy Accumulation

- Battery Voltage Increased: from 3.810 V to 3.823 V after 30-minute walk
- Conducted controlled discharge test using 220 Ω resistor
- Integrated energy calculated using: $P = V^2 / R$; $E = P \times t$
- Total Energy Measured: ~700 Joules

Conclusion

- Our 3D shoe sole prototype successfully capture mechanical energy from walking.
- A small voltage increase of about 0.009 V ~ 0.013 V (9 mV ~ 13 mV) indicates our piezoelectric system did indeed harvest and store a measurable amount of energy, albeit very small. The energy generated within 30 minutes of walking is enough to light a standard LED (3V, 20mA) for about 3 hours.

Brightness Level	Voltage	Current	Power
<i>Standard LED</i>	<i>3V</i>	<i>20 mA</i>	<i>0.06 W (60 mW)</i>

$$\text{Time} = 700\text{J} / 0.06\text{W} = 11666.67\text{seconds} \approx 194.4\text{minutes} \approx 3.24\text{hours}$$

- Harvested energy is small but can meaningfully power low-energy wearables, IoT sensors, or emergent lights in low-power applications.
- To power high-energy devices (like a smartphone which requires on average two watts of energy), longer activity time or parallel harvesting mechanisms are required.
- Future improvements of the 3D prototype can be done through: 1) Advanced piezo materials; 2) Optimized energy harvesting circuits.

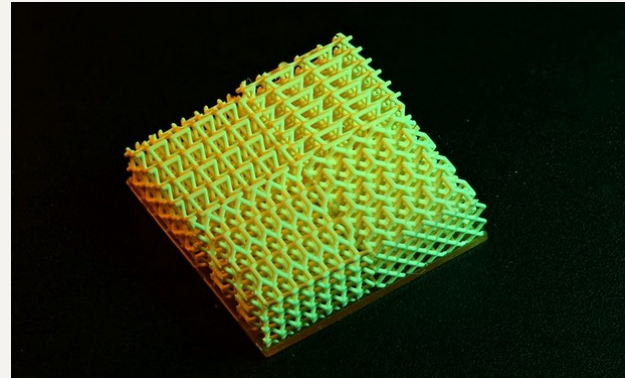
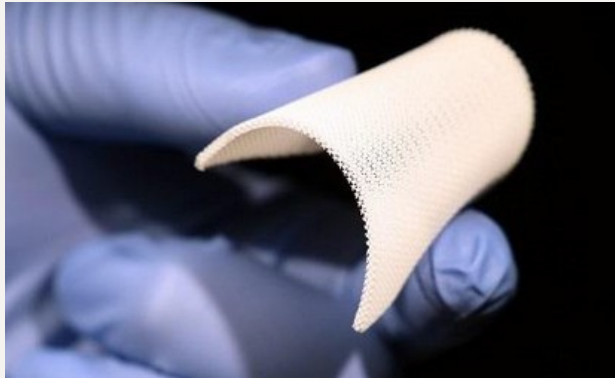
Trend In Real World

- **Pioneering Research:**

Dr. Rayne Zheng's research team at UC Berkeley is at the forefront of developing advanced **3D-printed piezoelectric materials**. <https://www.raynexzheng.com/>; <https://www.youtube.com/watch?v=HwpPpnEb4Ok>

- **Potential Collaboration:**

- Our project team is actively reaching out to Dr. Zheng's team for collaboration.
- We aim to gain expert insights, support, and potentially integrate cutting-edge methods from their groundbreaking research.



Major References

- <https://www.youtube.com/watch?v=PNjc9tUs96Y>
- <https://www.youtube.com/watch?v=FXyyuW4YoVU>
- <https://www.youtube.com/watch?v=zA0CCOYpnj4>
- <https://www.youtube.com/watch?v=p-yxiSrDiqQ>
- <https://www.youtube.com/watch?v=VcdIR3cuc0U>
- <https://www.youtube.com/watch?v=jYLG9XTBL94>
- <https://www.youtube.com/watch?v=svK6Gv7oM6U>